

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S5 (R,S)/S5 (PT) (R,S)/S5 (WP)(R)/S3 (PT)(S,FE) Examination November 2024 (2019 Scheme)

Course Code: HUT 310**Course Name: MANAGEMENT FOR ENGINEERS**

Max. Marks: 100

Duration: 3 Hours

*(Use of Normal Distribution Tables may be permitted)***PART A***(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|--|---|
| 1 | Explain the difference between management as a profession and management as an art | 3 |
| 2 | Describe the system approach to management and its key components. | 3 |
| 3 | Explain the concept of span of control and its significance in organizational design | 3 |
| 4 | Explain the role of leadership in directing organizational activities. | 3 |
| 5 | Define productivity and explain its measurement techniques | 3 |
| 6 | Explain the behavioural model of decision making and its significance | 3 |
| 7 | Describe the network construction process in project management | 3 |
| 8 | Discuss the importance of crashing in project management and its application | 3 |
| 9 | Outline the importance of human resource management in organizational success | 3 |
| 10 | Discuss the importance of corporate social responsibility in modern business. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|----|
| 11 | a) Explain the 14 principles of administrative management? | 14 |
| 12 | a) Explain Task and Responsibilities of a professional Manager. | 7 |
| | b) Explain the importance of management in an organisation | 7 |

Module -2

- | | | |
|----|---|---|
| 13 | a) Explain the mission, goals, strategy, programmes and procedures of an organisation with suitable examples. | 7 |
| | b) Illustrate the different types of organisation structures. | 7 |

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- 14 a) List the qualities expected of a leader. Discuss the different leadership styles and their impact on organizational performance. 7
- b) Differentiate between strategic and tactical planning. 7

Module -3

- 15 a) Explain different steps in decision tree analysis with an example. 7
- b) Explain in detail different productivity measurement indexes 7
- 16 a) For the payoff table given below, construct a decision tree and suggest the optimal decision which return maximum yield to the investor. 7

Investment option	Growing (₹)	Declining (₹)
Stocks	70	-13
Mutual funds	53	-5
Bonds	20	20
Probability	0.4	0.6

- b) Discuss controllable and non controllable factors affecting productivity 7

Module -4

- 17 a) The following details are available regarding a project. Determine the earliest and latest times, the total float for each activity, the critical activities and the project completion time. 14

Activity	Predecessor	Duration (Weeks)
A	-	12
B	A	7
C	A	11
D	A	8
E	A	6
F	B	10
G	C	9
H	D, F	14
I	E, G	13
J	H, I	16

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- 18 a) The following table shows the various activities for construction of a house. Find the critical path and optimal project duration. Also calculate total float for each activity. 14

Activity	Description	Immediate Predecessor	Duration (Weeks)
A	Site Preparation	-	2
B	Laying Foundation	A	6
C	Carpentry Work for Frames	-	4
D	Super-structure	B, C	12
E	Roofing	D	6
F	Electrification	E	4
G	Plumbing	E	3
H	Flooring and Plastering	F, G	10
I	Remaining Carpentry Works	H	5
J	Painting	I	8

Module -5

- 19 a) Discuss the importance of financial management in organizational success 7
b) Explain the marketing mix (4Ps) and its importance in marketing management 7
20 a) Elaborate different market segmentation strategies 7
b) What are intellectual property rights? Explain the business importance of patents. 7
